01 THEORY

Architecture of the CPU

The Central Processing Unit (CPU) processes instructions and data so that programs can run. It repeatedly fetches, decodes and executes instructions.

The fetch–decode–execute cycle

1 FETCH

Obtain the next instruction from main memory.

2 DECODE

The control unit interprets the instruction.

3 EXECUTE

Carry out the action, such as a calculation.

Why is this a cycle?

Once the instruction is executed, the processor continues with the next instruction. The stages repeat while the program runs.

One useful analogy

Think of a head chef: they collect an order (fetch), interpret what is requested (decode) and prepare the meal (execute). The analogy helps with sequence, but in an exam use the technical wording above.

What OCR expects you to know

- The purpose of the CPU and the three stages of the cycle.
- The functions of the ALU, control unit, cache and registers.
- What the MAR, MDR, Program Counter and Accumulator store - especially the difference between a memory address and data.

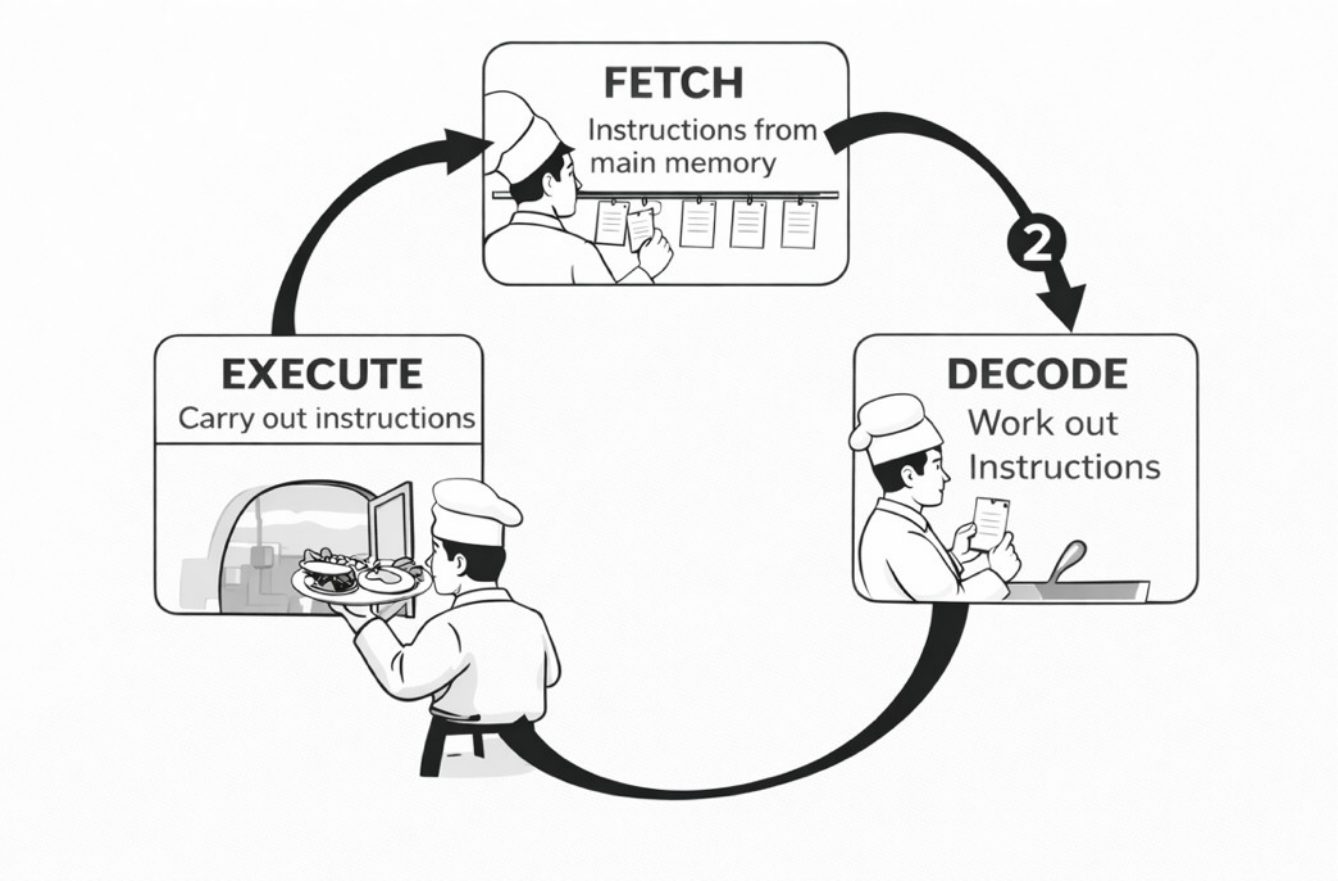
Specification boundary

J277 does not require a step-by-step account of transferring values between different registers at every stage of the cycle.

THEORY / VISUAL ANALOGY

The chef follows an order

Use the chef analogy to remember the stages. The explanation on the right gives accurate GCSE exam wording.



RESTAURANT ANALOGY		CORRECT EXAM WORDING
FETCH	Chef collects the next order.	The CPU fetches the next instruction from memory.
DECODE	Chef reads the order to understand it.	The Control Unit decodes the instruction.
EXECUTE	Chef carries out the order.	The CPU executes the instruction; the ALU may perform a calculation.

Remember: This is a repeating cycle, not a one-off task. Use the kitchen to recall the order; use fetch, decode, execute in your answer.

01 THEORY / FETCH–DECODE–EXECUTE

Three different instructions, three cycles

A chef does not cook the same order forever. They finish one instruction, then pick up the next, different instruction. The CPU follows the same repeating pattern.

INSTRUCTION 01 — **CHOP CARROTS**

FETCH	DECODE	EXECUTE
Chef picks up the order: CHOP CARROTS.	Chef works out what chopping carrots requires.	Chef chops the carrots.

NEXT INSTRUCTION

INSTRUCTION 02 — **STIR THE SOUP**

FETCH	DECODE	EXECUTE
Chef picks up a new order: STIR SOUP.	Chef works out what stirring the soup requires.	Chef stirs the soup.

NEXT INSTRUCTION

INSTRUCTION 03 — **PLATE THE MEAL**

FETCH	DECODE	EXECUTE
Chef picks up the next order: PLATE MEAL.	Chef works out how to plate the meal.	Chef plates the meal.

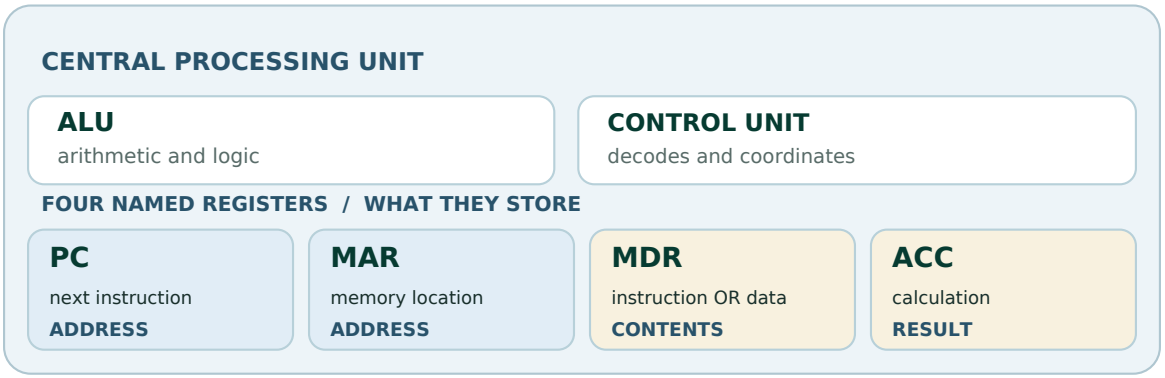
EXAM CONNECTION

For each instruction, the CPU fetches it from memory, the Control Unit decodes it, and the CPU executes it. It then fetches the next instruction. The stages repeat; the instruction changes.

01 THEORY

Components and registers

A CPU includes several parts with different purposes. Each contributes to handling instructions efficiently.



Core components

Arithmetic Logic Unit (ALU)
Performs arithmetic calculations and logical operations.

Control Unit (CU)
Decodes instructions and coordinates CPU activity using control signals.

Cache
Fast memory that holds frequently used data and instructions for quick access.

Registers
Very small, very fast storage locations for values currently needed by the CPU.

Von Neumann registers: address or data?

PC Program Counter
Address of the next instruction.

MAR Memory Address Register
Address of the memory location being accessed.

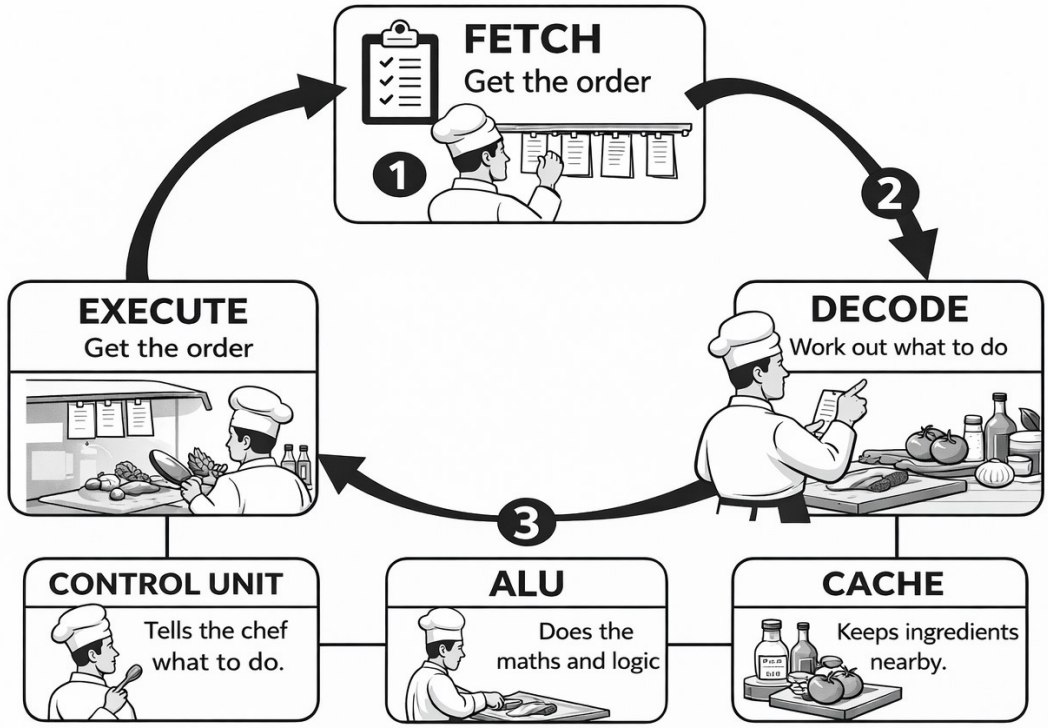
MDR Memory Data Register
Data or instruction being transferred to or from memory.

ACC Accumulator
Intermediate result of arithmetic or logic operations.

Remember: PC / MAR = addresses; MDR = instruction OR data; ACC = result.

01 THEORY / VISUAL ANALOGY The CPU works like a kitchen

The same familiar setting links the fetch–decode–execute cycle to the roles of the CPU components.

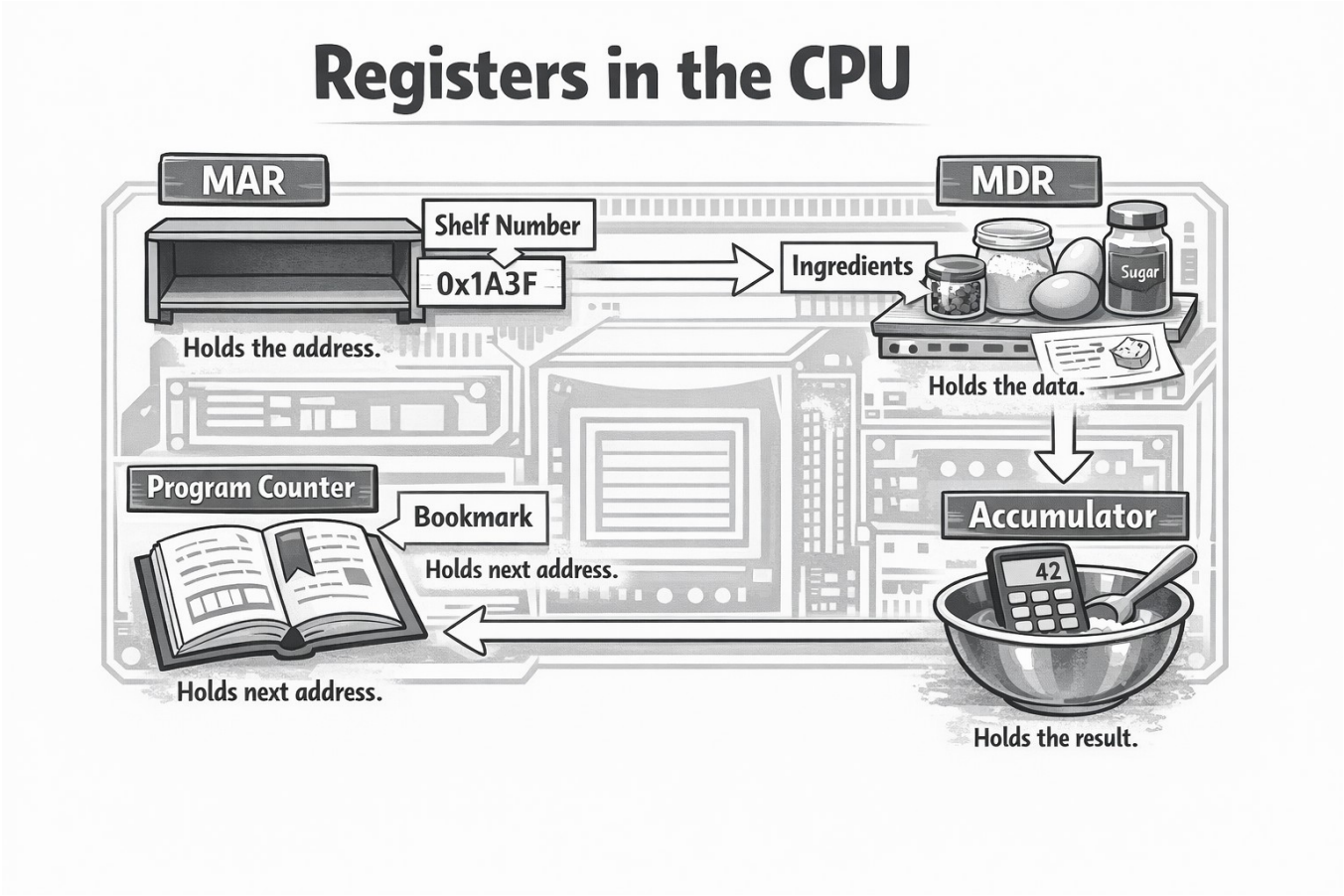


WHAT THE PICTURE REPRESENTS		CORRECT EXAM WORDING
CONTROL UNIT	Head chef directs the work.	Decodes instructions and coordinates operations using control signals.
ALU	Does the maths and logic needed.	Performs arithmetic operations and logical comparisons.
CACHE	Common ingredients kept close by.	Small, fast memory for frequently used data and instructions.

Exam tip: The analogy makes each role memorable; it does not replace the technical terms Control Unit, ALU and cache.

01 THEORY / VISUAL ANALOGY Registers: shelves and ingredients

The kitchen analogy distinguishes a memory address (where something is) from data (what it contains).



VISUAL CLUE		CORRECT EXAM WORDING
MAR	Shelf number identifies a location.	Memory Address Register stores the address of the memory location being accessed.
MDR	Ingredients are the actual contents.	Memory Data Register holds data or an instruction being transferred to or from memory.
PC	Bookmark shows where to read next.	Program Counter stores the address of the next instruction.
ACC	Mixing bowl holds the current result.	Accumulator stores an intermediate arithmetic or logical result.

02 KEY TERMINOLOGY

Words that earn marks

Use these terms accurately when you explain how the CPU works. The yellow highlighting is intentionally placed behind each keyword, as in the RHSC Computing lessons.

CPU
Processes data and executes instructions so that programs run.

Fetch
Retrieves the next instruction from main memory.

Decode
Interprets the instruction to identify the action needed.

Execute
Carries out the operation described by the instruction.

ALU
Performs arithmetic and logical operations.

Control Unit
Decodes instructions and coordinates CPU activities.

Cache
Small, fast memory holding frequently used instructions/data.

Register
Tiny, fast storage location within the processor.

Program Counter
Stores the address of the next instruction.

MAR
Stores the address of the memory location being accessed.

MDR
Stores data/instructions travelling to or from memory.

Accumulator
Stores intermediate results of calculations.

Exam reminder
For register questions, include the word address when describing the PC or MAR, and data/instruction when describing the MDR.

03 WORKED EXAMPLES

See how marks are earned

WORKED EXAMPLE 1

Question: Explain the three stages of the fetch–decode–execute cycle. [3 marks]

FETCH The CPU retrieves the next instruction from memory.

DECODE The control unit interprets the instruction.

EXECUTE The CPU carries out the required instruction, such as an ALU calculation.

Why this earns 3 marks

Each distinct stage has the correct action. The answer uses precise exam language rather than relying on the chef analogy.

WORKED EXAMPLE 2

Question: A program is about to access memory location 120. Which register holds the location address, and which holds the data read from that location? [2 marks]

MAR Holds the address 120 - the location to access.

MDR Holds the actual data or instruction read from that location.

The distinction

The MAR stores where; the MDR stores what. You do not need to memorise the sequence of data passing between registers for this specification point.

04 EXAM PRACTICE

Try it yourself · 1 of 5

Answer the questions before looking at the separate mark scheme. Marks are shown beside each question.
Attempt each question before checking the separate mark scheme.

Q1 1 MARK

What is the purpose of the CPU?

Q2 1 MARK

Name one component of the CPU.

Q3 1 MARK

What does the Program Counter store?

Q4 2 MARKS

State two roles of the Control Unit.

04 EXAM PRACTICE

Try it yourself · 2 of 5

Answer the questions before looking at the separate mark scheme. Marks are shown beside each question.
Attempt each question before checking the separate mark scheme.

Q5

2 MARKS

State two types of operation that the ALU performs.

Q6

2 MARKS

Match each CPU component to its function. Connect ALU, Control Unit and Registers to the correct functions.

Choose from:

A holds data and addresses being used

B decodes instructions and controls operations

C performs calculations and logic

ALU	Write A, B or C: _____
Control Unit	Write A, B or C: _____
Registers	Write A, B or C: _____

Q7

3 MARKS

Explain the Fetch–Decode–Execute cycle.

04 EXAM PRACTICE

Try it yourself · 3 of 5

Answer the questions before looking at the separate mark scheme. Marks are shown beside each question.
Attempt each question before checking the separate mark scheme.

Q8

3 MARKS

Explain why registers are used inside the CPU.

Q9

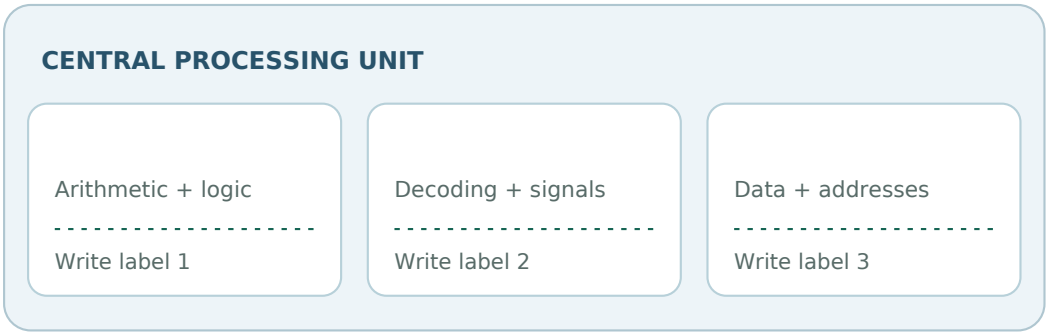
3 MARKS

Explain the role of cache memory in the CPU.

Q10

3 MARKS

Label the ALU, Control Unit and registers on the blank CPU diagram below.



04 EXAM PRACTICE

Try it yourself · 4 of 5

Answer the questions before looking at the separate mark scheme. Marks are shown beside each question.
Attempt each question before checking the separate mark scheme.

Q11 4 MARKS

Describe the role of the ALU and the Control Unit.

Q12 4 MARKS

Describe the role of the MAR and MDR.

04 EXAM PRACTICE

Try it yourself · 5 of 5

Answer the questions before looking at the separate mark scheme. Marks are shown beside each question.
Attempt each question before checking the separate mark scheme.

Q13 4 MARKS

Compare the MAR and MDR.

Q14 6 MARKS

Explain how the CPU processes instructions using the Fetch–Decode–Execute cycle.

05 ANSWERS & MARK SCHEME

Check your work · 1 of 4

Read your answer before self-marking. Award marks for correct computing ideas. These are indicative model answers, not an official OCR mark scheme.

Q1 · 1 mark

The CPU processes data and instructions so that programs can run.

Q2 · 1 mark

Arithmetic Logic Unit (ALU), Control Unit (CU), cache or registers.

Q3 · 1 mark

The memory address of the next instruction to be fetched.

Q4 · 2 marks

It decodes instructions and sends control signals to coordinate CPU operations.

- 1 mark: decodes instructions.
- 1 mark: coordinates/controls execution, e.g. by sending control signals.

Q5 · 2 marks

Arithmetic operations (for example, addition) and logical operations (for example, AND).

- 1 mark: arithmetic.
- 1 mark: logical operations.

05 ANSWERS & MARK SCHEME

Check your work · 2 of 4

Read your answer before self-marking. Award marks for correct computing ideas. These are indicative model answers, not an official OCR mark scheme.

Q6 · 2 marks

ALU → calculations; Control Unit → decodes/coordinates; Registers → small, fast storage for current data and addresses.

Q7 · 3 marks

The CPU fetches the next instruction from memory. The Control Unit decodes it to determine what action is required. The CPU then executes it, possibly using the ALU.

- 1 mark: fetches next instruction from memory.
- 1 mark: decodes/interprets the instruction.
- 1 mark: carries out/executes the instruction.

Q8 · 3 marks

Registers are small, very fast storage locations within the CPU. They hold addresses, data or results currently being used, giving the processor rapid access during instruction processing.

- 1 mark: small storage in/very close to CPU.
- 1 mark: holds data, addresses or results in current use.
- 1 mark: faster access supports efficient processing.

Q9 · 3 marks

Cache is small, fast memory close to or within the CPU. It holds frequently used data and instructions. This reduces time spent accessing the slower main memory.

- 1 mark: small, fast memory.
- 1 mark: frequently used data/instructions.
- 1 mark: quicker access/reduced RAM accesses.

05 ANSWERS & MARK SCHEME

Check your work · 3 of 4

Read your answer before self-marking. Award marks for correct computing ideas. These are indicative model answers, not an official OCR mark scheme.

Q10 · 3 marks

From left to right: ALU; Control Unit; Registers.

Q11 · 4 marks

The ALU performs arithmetic operations such as addition and logical operations such as comparisons. The Control Unit decodes instructions and coordinates how CPU components act by sending control signals.

- 1 mark: ALU arithmetic.
- 1 mark: ALU logic.
- 1 mark: CU decodes.
- 1 mark: CU controls or coordinates instructions/CPU operations.

Q12 · 4 marks

The MAR holds the address of the memory location being accessed. The MDR holds the data or instruction being transferred to or from memory. They are both CPU registers used in memory access.

- 1 mark: MAR identified as a register.
- 1 mark: MAR stores memory address.
- 1 mark: MDR identified as a register.
- 1 mark: MDR stores data/instruction being read or written.

05 ANSWERS & MARK SCHEME

Check your work · 4 of 4

Read your answer before self-marking. Award marks for correct computing ideas. These are indicative model answers, not an official OCR mark scheme.

Q13 · 4 marks

Both are registers used when the CPU accesses memory. The MAR holds a location address - it identifies where the data is stored. The MDR holds the actual data or instruction being read from or written to that location.

- 1 mark: both are registers / involved in memory access.
- 1 mark: MAR stores an address.
- 1 mark: MDR stores data or an instruction.
- 1 mark: explicit comparison of location vs content.

Q14 · 6 marks

The CPU repeats the fetch–decode–execute cycle to run a program. During fetch, it obtains the next instruction from memory. The Control Unit decodes the instruction to determine the required action. During execute, the CPU carries out that action; for example, the ALU might perform an arithmetic or logic operation. The Program Counter identifies the address of the next instruction, while registers provide fast storage for addresses, data and results. The Accumulator may hold the result of an ALU operation.

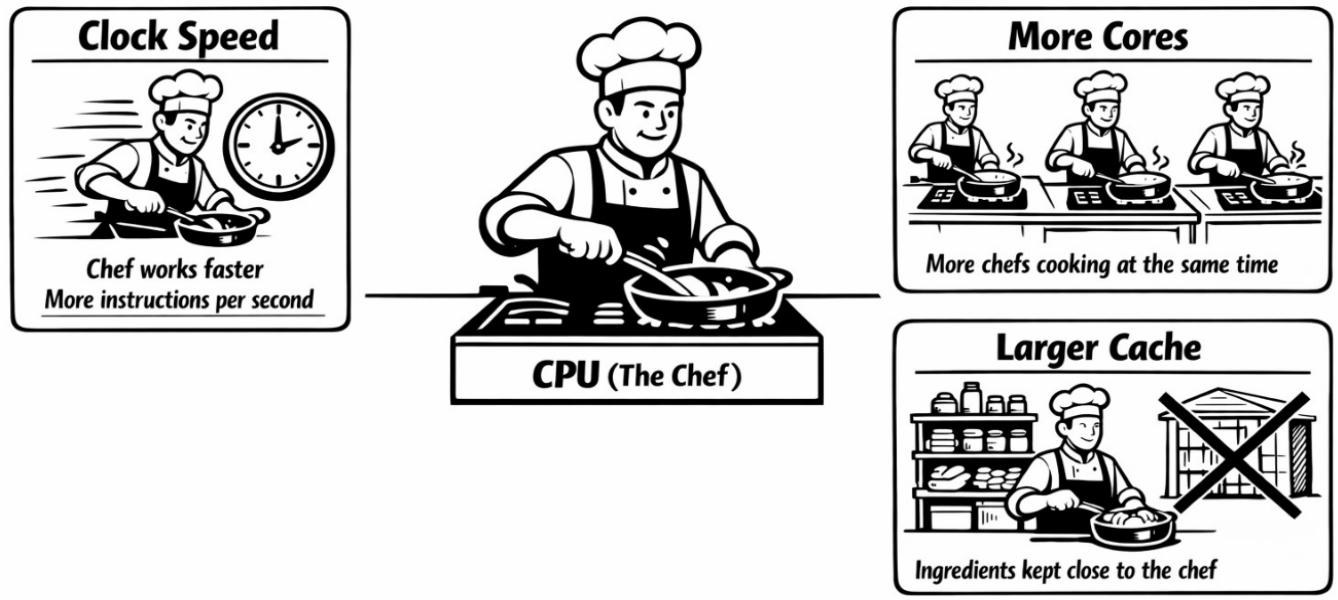
- 1 mark: cycle repeats to process a program.
- 1 mark: fetches instruction from memory.
- 1 mark: decodes/understands instruction.
- 1 mark: executes the required action.
- 1 mark: correct component role (e.g., CU or ALU).
- 1 mark: correct register role (PC, MAR, MDR or Accumulator).

01 THEORY

CPU performance

Three connected ideas: clock speed, cache size and number of cores.

One familiar kitchen, three ways to improve performance



Link the picture to the technical explanation

1 **Clock speed**

The chef works faster → more clock cycles per second.

2 **Number of cores**

More chefs can work at the same time → parallel work.

3 **Cache size**

Ingredients are kept close → less waiting for frequently used data.

Exam reminder

A faster clock, larger cache and more cores may improve performance, but the effect depends on the workload. More cores help most when tasks can run in parallel.

01 THEORY

See the image. Explain the CPU.

The drawings are your memory cues; the right-hand wording earns marks.



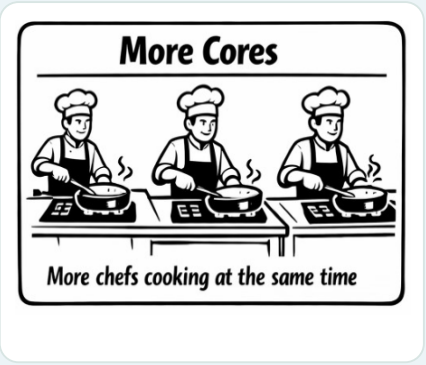
Chef works faster

CLOCK SPEED

TECHNICAL EXPLANATION

Clock speed is the rate of CPU clock cycles, measured in Hz or GHz. Raising it can allow the CPU to process more instructions in a given time.

Remember: Do not assume one whole instruction is completed on every clock tick.



More chefs together

NUMBER OF CORES

TECHNICAL EXPLANATION

Each CPU core can process instructions. More cores can carry out work in parallel and improve multitasking when tasks can use them.

Remember: One task cannot automatically use every core.



Ingredients kept close

CACHE SIZE

TECHNICAL EXPLANATION

Cache is small, fast memory holding frequently used data and instructions. A larger cache may reduce slower access to RAM.

Remember: More cache helps when the CPU can reuse the stored items.

01 THEORY

Using factors in combination

An exam might ask which CPU is better for a particular job. Explain why.

Compare two different processors

PROCESSOR A		PROCESSOR B	
Clock speed	**3.6 GHz**	Clock speed	**3.1 GHz**
Number of cores	**2 cores**	Number of cores	**6 cores**
Cache	**4 MB**	Cache	**12 MB**

Which would you choose?

Several independent programs running together

Processor B may be better: six cores offer more parallel capacity, and its larger cache may reduce memory delays. A has a higher clock speed, so it might perform better on some tasks that mainly use one core.

One task that cannot run in parallel

Processor A might be faster because of its higher clock speed, but GHz alone cannot prove that one entire processor is faster. The program and other factors matter.

Strong exam answer

Name the factor → explain how it changes CPU operation → link it to the task. For comparisons, explain a trade-off rather than claiming one CPU is always best.

02 KEY TERMINOLOGY

Words that earn marks

These terms connect your chef analogy to the correct OCR terminology.

CPU performance

How efficiently a processor handles work, including speed of execution.

Clock speed

The frequency of a CPU clock, measured in hertz (Hz).

Hertz (Hz)

One clock cycle per second; GHz means billions of cycles per second.

Cache

Small, fast memory storing frequently used data or instructions.

Cache size

The amount of data/instructions that can be held in cache.

CPU core

A processing unit within a processor that can execute instructions.

Parallel processing

Executing different parts of work at the same time.

Multitasking

Making progress on more than one task; multiple cores can help.

Tip: using the correct terms helps your explanation earn marks.

03 WORKED EXAMPLES

See how marks are earned

Follow the reasoning, then use the same approach for independent practice.

WORKED EXAMPLE 1 · 3 marks

Explain how increasing cache size can improve CPU performance.

1 State what cache does

Cache holds frequently used data and instructions.

2 Explain the speed advantage

The CPU accesses cache faster than it accesses RAM.

3 Link to performance

A larger cache can reduce time waiting for data, improving speed.

WORKED EXAMPLE 2 · 4 marks

A computer runs a web browser, a video call and a game together. Explain how four CPU cores may help compared with two.

1. More cores can process different instructions or tasks at the same time.
2. The browser, video call and game may make progress in parallel.
3. This can improve responsiveness and reduce waiting between tasks.
4. The benefit depends on whether the programs can use multiple cores.

Why these examples earn marks: each point explains a technical idea and connects it to the requested performance effect.

04 EXAM PRACTICE

Your turn · 1 of 4

Attempt these questions before checking the answers. Write enough for the marks shown.

Q1 · 1 mark · Define

What is meant by CPU performance?

Q2 · 1 mark · Define

Name one factor that affects CPU performance.

Q3 · 1 mark · Define

What is clock speed measured in?

04 EXAM PRACTICE

Your turn · 2 of 4

Attempt these questions before checking the answers. Write enough for the marks shown.

Q4 · 2 marks · Explain

State two ways increasing clock speed improves CPU performance.

Q5 · 2 marks · Explain

State two benefits of having multiple CPU cores.

Q6 · 3 marks · Explain

Explain how clock speed affects CPU performance.

04 EXAM PRACTICE

Your turn · 3 of 4

Attempt these questions before checking the answers. Write enough for the marks shown.

Q7 · 3 marks · Explain

Explain the role of cache memory in improving CPU performance.

Q8 · 3 marks · Explain

Explain how the number of CPU cores affects performance.

Q9 · 4 marks · Describe roles

Describe two factors that affect CPU performance.

04 EXAM PRACTICE

Your turn · 4 of 4

Attempt these questions before checking the answers. Write enough for the marks shown.

Q10 · 4 marks · Describe roles

Describe how cache size and number of cores improve CPU performance.

Q11 · 6 marks · Discuss / Explain

Explain how clock speed, cache size and number of cores affect CPU performance.

05 ANSWERS & MARK SCHEME

Check your work · 1 of 3

Mark the scientific idea, not only the exact wording. Use the model answer to improve your own.

Q1 · 1 mark

CPU performance is how quickly and efficiently a CPU handles instructions and work.

Marking points

Accurate definition mentioning the speed/efficiency of processing.

Q2 · 1 mark

Clock speed; cache size; or number of CPU cores.

Marking points

Any one valid listed characteristic.

Q3 · 1 mark

Hertz (Hz).

Marking points

Hz, kHz, MHz or GHz as a suitable frequency unit.

Q4 · 2 marks

Higher clock speed gives more clock cycles each second; this can allow instructions to be processed more quickly.

Marking points

One mark for more cycles per second; one for linked effect on speed/work completed.

05 ANSWERS & MARK SCHEME

Check your work · 2 of 3

Mark the scientific idea, not only the exact wording. Use the model answer to improve your own.

Q5 · 2 marks

Multiple cores can work on different tasks simultaneously; this can improve multitasking or responsiveness when software can run in parallel.

Marking points

One mark for parallel work; one for an explained performance/multitasking benefit.

Q6 · 3 marks

Clock speed is the rate of clock cycles per second, measured in Hz. A higher clock rate gives the processor more cycles in a given time, which can allow more instructions to be executed and reduce processing time.

Marking points

Clock frequency; change from increasing it; linked effect on performance.

Q7 · 3 marks

Cache holds frequently used data and instructions close to the CPU. It is faster to access than RAM, so it reduces delays fetching those items and may speed up processing.

Marking points

What cache holds; faster than RAM; fewer delays/more performance.

Q8 · 3 marks

Each CPU core can process instructions. With more cores, suitable tasks can run in parallel, helping the computer process multiple tasks at once and improve responsiveness.

Marking points

Meaning of cores; parallel operation; effect on suitable workload.

05 ANSWERS & MARK SCHEME

Check your work · 3 of 3

Mark the scientific idea, not only the exact wording. Use the model answer to improve your own.

Q9 · 4 marks

Clock speed: a higher rate provides more CPU cycles each second, so processing may complete sooner. Cache size: a larger cache can hold more frequently used data, reducing access to slower RAM.

Marking points

For each of two relevant factors, credit the characteristic and a valid explained effect.

Q10 · 4 marks

Larger cache means more frequently used information may be available quickly rather than retrieved from RAM, reducing waiting. Additional cores allow compatible tasks to be processed simultaneously, helping multitasking.

Marking points

Two linked explanatory points for cache and two for cores, as appropriate.

Q11 · 6 marks

Clock speed controls how many clock cycles occur per second; increasing it can allow instructions to be processed faster. A larger cache stores more frequently used data/instructions in fast memory and reduces slower RAM accesses. More CPU cores allow suitable work to proceed in parallel, improving performance when several tasks run. Actual improvement depends on how the program uses the CPU.

Marking points

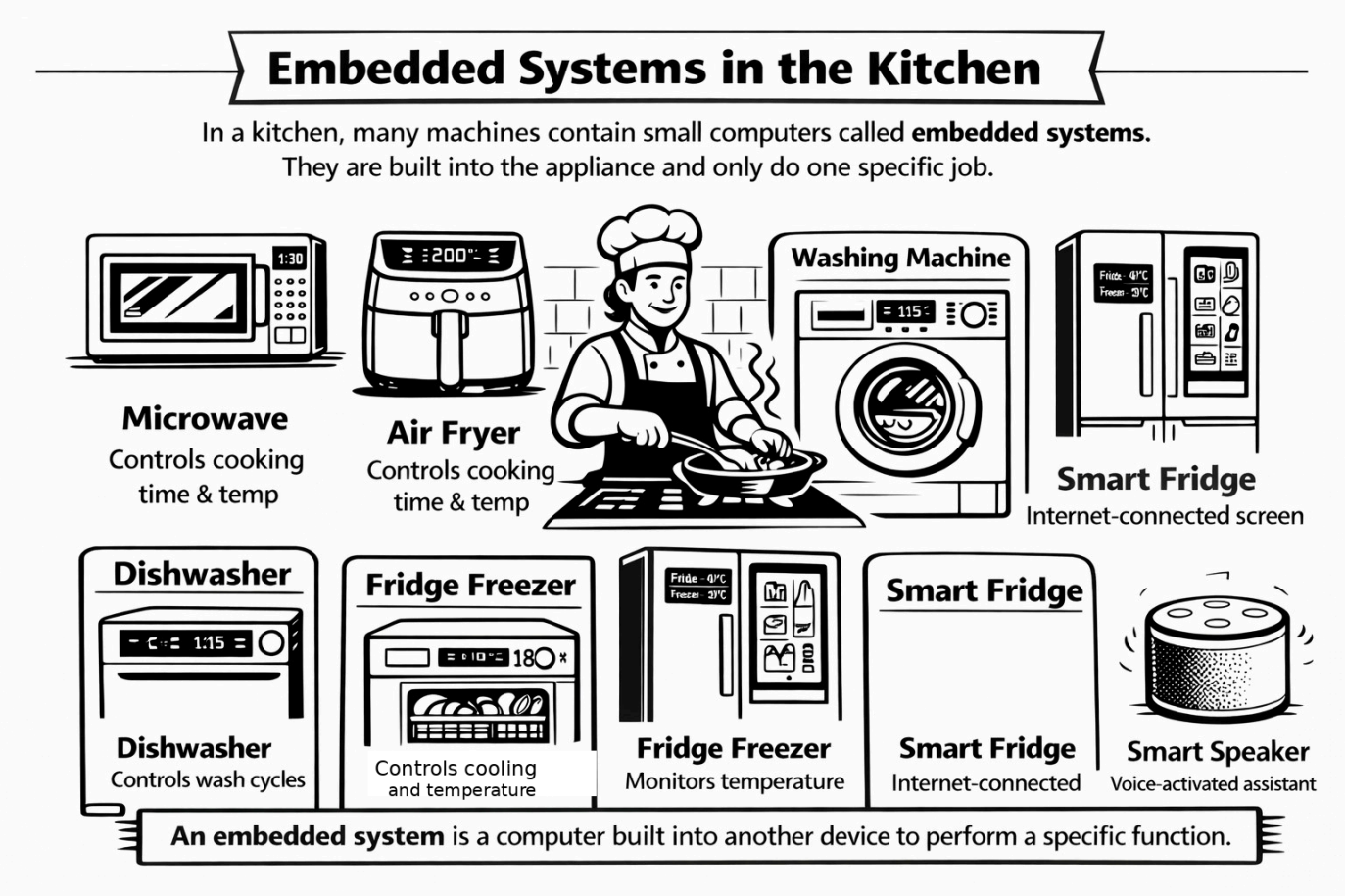
Indicative: explain clock speed and its effect; cache and its effect; cores and parallel work; link to overall performance and/or suitability of workload.

01 THEORY

Embedded systems in the kitchen

A familiar appliance can contain its own small, task-focused computer.

Embedded systems in the kitchen



Embedded system

A computer built into another device to control or monitor a dedicated function. It is not normally used like a general-purpose laptop.

Read the image carefully

Examples include a microwave managing heating time, a dishwasher controlling a wash cycle and a fridge controller monitoring temperature.

A fridge-freezer controller monitors the temperature and switches cooling on when needed.

01 THEORY

From appliance to exam answer

Three examples from the kitchen illustration.



MICROWAVE

INPUT

User sets cooking time

PROCESS

Controller counts down

OUTPUT

Turns heating off at the end



WASHING MACHINE

INPUT

User chooses wash cycle

PROCESS

Controller follows programme

OUTPUT

Operates pump, drum, heater



FRIDGE CONTROLLER

INPUT

Temperature sensor input

PROCESS

Controller compares to target

OUTPUT

Switches cooling as needed

A controller receives inputs, follows instructions and controls outputs.

01 THEORY

What makes them different?

Compare the function, interface and requirements of different devices.

Embedded

A washing-machine controller has a defined task. It reads inputs and controls a specific appliance.

Designed for a dedicated role

General-purpose

A laptop can run many different applications chosen by its user: documents, games, videos and coding.

Runs a wide range of programs

Typical characteristics — avoid overgeneralising

Designed for a specific role

Works within an appliance or a larger system.

Often a limited interface

Could have buttons, a small display or no screen.

Often reliable, low power

Requirements depend on the device and its environment.

Some are real-time

A response must meet a required time deadline.

Uses memory and firmware

Programs and temporary data need storage; memory sizes vary.

Exam habit: say “often” or “typically” unless the question specifies an absolute.

02 KEY TERMINOLOGY

Words behind the kitchen analogy

The exact lesson highlighter marks words used in exam explanations.

Embedded system

A computer integrated into a larger device for a dedicated task.

Dedicated function

The particular job or limited set of jobs a controller is designed to perform.

Controller

Processes inputs and issues outputs to manage an appliance.

Sensor

Detects a physical condition, such as temperature.

Input

A signal or information received by a system.

Output

An action or signal generated by the controller.

Real-time

Must respond within a specified deadline when timing matters.

User interface

The way a user operates a device or receives information.

Firmware / ROM

Program instructions kept in non-volatile storage (where appropriate).

RAM

Working memory for temporary values used while the system runs.

Exam precision: ROM and RAM capacity vary; not every embedded system is real-time.

03 WORKED EXAMPLES

Example 1 · Washing machine

Follow input → process → output, then turn the diagram into an answer.

Exam-style example (4 marks)

Explain how a washing machine can use an embedded system to control a wash cycle.

INPUT

User selects a wash programme; sensors may report water level.

▼

PROCESS

Embedded controller follows the selected sequence and sensor readings.

▼

OUTPUT

Motor, valves, pump or heater are switched on or off.

Model answer (4 linked points)

The washing machine contains a computer controller built into the appliance [1]. A user selects a wash programme [1]. The controller receives inputs such as sensor readings and follows the programmed sequence [1]. It controls outputs, for example the pump or motor, to carry out the cycle [1].

03 WORKED EXAMPLES

Example 2 · Microwave and laptop

Apply the characteristics without claiming that every device behaves identically.

Exam-style example (3 marks)

Explain why a microwave controller is an embedded system but a laptop is general-purpose.

1 Identify the system

The microwave contains a computer built inside another device.

2 Explain its purpose

It has a dedicated role: timing and controlling microwave operation.

3 Compare

A laptop allows users to run many unrelated programs.

Check the common misconceptions

Real-time does not mean “instant”: it means meeting a deadline when the application requires it.

ROM and RAM are not guaranteed to be “large” or “small”. Firmware may be held in non-volatile storage and RAM may hold temporary working data; the precise amounts depend on the device.

04 EXAM PRACTICE

Questions 1–5

Attempt all questions before looking at the answers.

Q1 1 mark

What is an embedded system?

Q2 1 mark

Name one characteristic of an embedded system.

Q3 1 mark

Where is the software for an embedded system usually stored?

Q4 2 marks

State two characteristics of an embedded system.

Q5 2 marks

State two reasons embedded systems use low power.

04 EXAM PRACTICE

Questions 6–8

Explain your reasons clearly; use the correct technical terms.

Q6 3 marks

Explain what is meant by a real-time embedded system.

Q7 3 marks

Explain why embedded systems have a limited user interface.

Q8 3 marks

Explain why embedded systems use a large ROM and a small amount of RAM.

04 EXAM PRACTICE

Questions 9–11

Use examples and connect characteristics to their purpose.

Q9 4 marks

Describe two characteristics of embedded systems.

Q10 4 marks

Describe how real-time operation and low power consumption are important in embedded systems.

Q11 6 marks

Explain the main characteristics of embedded systems.

05 ANSWERS & MARK SCHEME

Questions 1–5

Use these marking points to check and improve your answers.

Q1 · 1 marks

What is an embedded system?

- Built into another device [1].
- Performs a dedicated or specific function (helpful extra detail).

Q2 · 1 marks

Name one characteristic of an embedded system.

- Any one typical characteristic: dedicated purpose / built into larger device / restricted interface / power-efficient design / time-constrained operation where appropriate [1].

Q3 · 1 marks

Where is the software for an embedded system usually stored?

- ROM or another suitable non-volatile storage holding firmware [1].

Q4 · 2 marks

State two characteristics of an embedded system.

- Built into a larger system [1].
- Designed for a particular task or limited set of tasks [1].

Q5 · 2 marks

State two reasons embedded systems use low power.

- Lower consumption helps extend battery life in portable devices [1].
- Lower energy/heat demand benefits always-on or constrained appliances [1].

05 ANSWERS & MARK SCHEME

Questions 6–8

Use these marking points to check and improve your answers.

Q6 · 3 marks

Explain what is meant by a real-time embedded system.

- Timing of the response affects whether operation is correct [1].
- The controller responds to an event or input [1].
- It must complete the response within a specified deadline [1].

Q7 · 3 marks

Explain why embedded systems have a limited user interface.

- The system serves a limited range of user tasks [1].
- Many controls can therefore be pre-programmed or automatic [1].
- A few controls or indicators may be sufficient, depending on the appliance [1].

Q8 · 3 marks

Explain why embedded systems use a large ROM and a small amount of RAM.

- Firmware must be retained when power is removed, so it may be kept in ROM or other non-volatile storage [1].
- RAM stores temporary data and working values during operation [1].
- Memory is provisioned to suit the task; not every embedded system has “large ROM and small RAM” [1].

Self-mark and correct

Tick the correct points you made. Underline technical terms you missed, then improve one answer.

05 ANSWERS & MARK SCHEME

Questions 9–11

Use these marking points to check and improve your answers.

Q9 · 4 marks

Describe two characteristics of embedded systems.

- Embedded in a larger device [1] and carries out a dedicated task [1].
- Further characteristic with explanation (e.g. restricted interface because controls are task-specific) [1+1].

Q10 · 4 marks

Describe how real-time operation and low power consumption are important in embedded systems.

- A time-critical controller must complete a required response within a specified deadline [1], supporting safe or correct operation [1].
- Power-efficient design helps battery-powered or always-on devices run longer [1] and reduce energy or heat demands [1].

Q11 · 6 marks

Explain the main characteristics of embedded systems.

- Embedded in a larger system, such as a washing machine [1].
- Designed for a dedicated control or monitoring role [1].
- Usually uses task-specific hardware or firmware [1].
- May use a limited interface because only relevant controls are needed [1].
- Some are real-time, meeting response deadlines [1].
- Often optimised for cost, size and power needs rather than general-purpose computing [1].